

Design and Experimental Characterisation of a 2 Mbit/s Polymer Optical Fibre Link

Automatic Gain Control, Hysteresis Detection and Signal Integrity

Yiling Fan

Department of Engineering, University of Cambridge

Technical report based on a collaborative engineering project completed as part of the Cambridge Engineering Tripos.

Abstract

This project presents the design, implementation and characterization of a 2 Mbit/s polymer optical fibre communication link. The complete system consists of a pseudo-random bit sequence generator, LED transmitter, polymer optical fibre channel, photodiode receiver, automatic gain control (AGC) circuitry and a high-speed comparator-based decision stage. System-level analysis was performed through power budget calculations and experimental eye-diagram measurements. A measured link attenuation of approximately 13 dB was successfully compensated by the receiver chain, enabling reliable digital data recovery over a 20 m fibre channel. The comparator achieved logic-level restoration with switching times below the required 200 ns specification, while the AGC stage improved tolerance to variations in received optical power. Eye-diagram characterization at 2 Mbit/s demonstrated a large noise margin and low timing jitter, confirming robust link operation. The successful transmission of real CD audio data further validated the practical performance of the completed communication system.

1 Introduction

Optical fibre communication infrastructures represent a critical backbone for high-speed, noise-immune data transfer in modern industrial and local area networks (LANs). This project focuses on the end-to-end design, hardware implementation, and empirical characterization of a complete electro-optical communication link. The primary technical milestone is to achieve synchronous, real-time digital transmission at a baseline rate of 2 Mbit/s over a Polymer Optical Fibre (POF) medium using a visible red LED transmitter ($\lambda \approx 650$ nm).

In addition to PRBS testing, a CD audio data stream was transmitted through the link and recovered for playback. This provided a practical system demonstration alongside electrical and optical measurements.

2 System Architecture and Link Budget

2.1 Project Specifications and Design Challenges

The primary engineering challenge arises from the wide dynamic range and non-ideal attenuation characteristics of the physical 20 m polymer optical fibre (POF) channel. Measured link losses of up to approximately 13 dB, together with variations in coupling efficiency, fibre bending and ambient light conditions, can produce substantial fluctuations in the received optical power. Without appropriate signal conditioning, these variations may lead to two undesirable operating regimes:

- **Under-drive:** Excessive attenuation reduces the received signal amplitude below the comparator threshold, increasing the probability of incorrect logic decisions and ultimately causing link failure.
- **Over-drive:** Excessive optical power can drive the receiver amplification stages towards saturation, resulting in output clipping, waveform distortion and reduced signal fidelity.

Both conditions reduce the available system margin and degrade the reliability of digital data recovery. To address these challenges, the receiver was designed around a high-speed AD8055-based amplification chain followed by a comparator decision stage. In addition, an Automatic Gain Control (AGC)

architecture was investigated to improve tolerance to variations in received optical power and maintain a more consistent signal amplitude at the receiver output.

2.2 Global Link Architecture and Signal Flow

To validate the communication infrastructure, the global link was engineered as a continuous, end-to-end electro-optical system. The systematic signal propagation translates data from the consumer audio source, through the physical photonic channel, and back into processed digital logic to drive the speaker output. The holistic hardware architecture and cascading signal flow are structurally mapped in Figure 1.

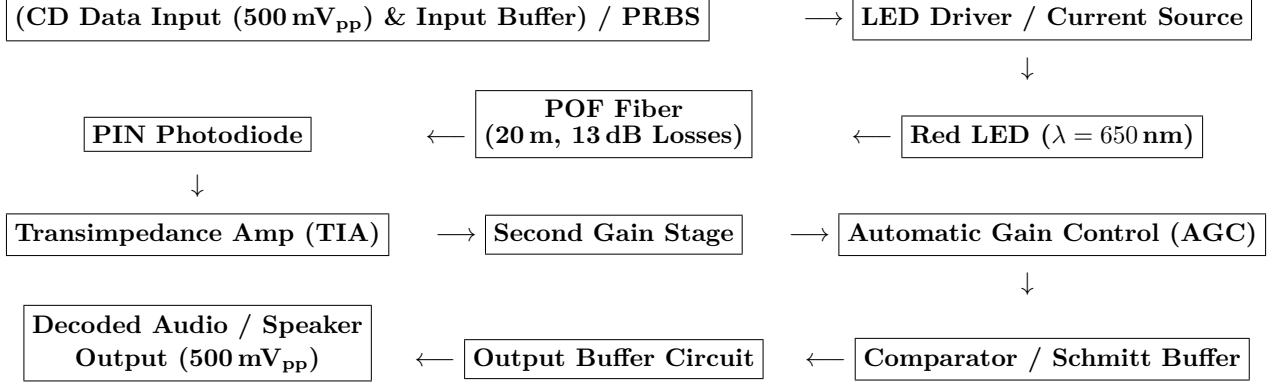


Figure 1: Functional Block Diagram of the End-to-End Optical Fibre Communication Link Transmitting CD Audio to a Speaker Output Destination.

The diagram summarises the physical link and audio demonstration. The comparator restores digital logic; audio playback also requires the downstream decoding interface. The AGC control-voltage circuitry and MOSFET attenuator condition the received signal before digital recovery.

To see the detailed circuit diagrams for each functional block—including the transmitter, receiver, AGC, header detection, and clock recovery stages—please refer to **Appendix A**.

2.3 Optical Link Power Budget Analysis

To theoretically validate link feasibility before individual sub-system optimization, a streamlined optical power budget was evaluated during the collaborative group phase, using measured transmitter and receiver parameters.

The transmitting Red LED injects a measured peak optical power of $P_{tx} = 6 \mu\text{W}$ (-22.2 dBm) into the POF core. At the receiver front-end, combining the PIN photodiode responsivity ($R_\lambda = 0.41 \mu\text{A}/\mu\text{W}$) with the total conversion gain of the TIA and secondary amplification stages ($A_{total} = 4.18 \times 10^6 \Omega$) yields an integrated optical-to-electrical system sensitivity of $S_{sys} = 1714 \text{ mV}/\mu\text{W}$.

To securely cross the fixed 300 mV comparator threshold (V_{th}), the receiver dictates a minimum required optical power of $P_{rx, min} \approx 0.175 \mu\text{W}$ (-37.6 dBm), establishing a maximum allowable system budget of 15.4 dB. Accounting for a cumulative path attenuation of 14.0 dB (comprising 12.0 dB bulk POF loss over 20 m and 2.0 dB coupling insertion loss), the link secures a positive net safety margin of:

$$\text{Margin}_{net} = 15.4 \text{ dB} - 14.0 \text{ dB} = +1.4 \text{ dB}$$

The estimated +1.4 dB margin supports nominal link feasibility. It is a budget estimate rather than a guarantee under component, coupling or noise variations. The 300 mV threshold belongs to this budget estimate; the later comparator section documents the implemented biased thresholds.

3 Individual Technical Contribution

To guarantee parallel engineering development and systematic optimization under the unified team designation **Bandwidth Bandits**, technical responsibilities were distributed structurally among the group members: Yizhen Wang, Tianze Li, and Yiling Fan (the author).

The initial phases of the project were executed as a collective Group Work effort. This collaborative baseline included the empirical characterization of the non-ideal 20 m POF channel, the mathematical calculation of the optical link budget, the block diagram, and the end-to-end integration of the complete audio transmission pipeline.

Following the establishment of the core system topology, individual ownership was assigned to streamline development, while maintaining continuous cross-functional technical support and collaborative hardware debugging across all interfaces:

- **Yizhen Wang:** Led the synthesis, simulation, and hardware implementation of the Transmitter modulation circuitry, the PRBS data generation network, and the digital Header Detection circuit.
- **Tianze Li:** Led the design, optimization, and validation of the receiver front-end topologies, alongside the complex dynamic Clock Recovery circuit.
- **Yiling Fan (Author):** Assumed primary technical ownership for the prototyping, characterization, and troubleshooting of the Potential Divider-Based Feedforward AGC Linear Amplification Infrastructure and the subsequent Precision Hysteresis Comparator Stage.

While each sub-system was spearheaded by its respective owner, the physical debugging on the prototyping matrix and the final system integration were highly collaborative endeavors, with all members actively assisting in end-to-end link verification. The subsequent individual analysis chapters reflect the engineering choices, non-ideal physical troubleshooting, and verification strategies executed primarily within the author’s scope of ownership.

3.1 System Integration and Test Methodology

To minimise debugging complexity, the communication link was integrated incrementally. Individual subsystems were first verified independently before being combined into progressively larger functional blocks. The transmitter and PRBS generator were initially characterised using direct electrical measurements. The receiver front-end was subsequently validated using controlled electrical test signals prior to optical testing.

Following successful subsystem verification, the transmitter and receiver were connected through both short and 20 m polymer optical fibre links. Comparator threshold settings and receiver gain were adjusted to achieve reliable digital recovery under varying optical attenuation conditions. Finally, the complete system was evaluated using eye-diagram measurements and real CD audio transmission to verify end-to-end performance at the target data rate.

4 Automatic Gain Control

4.1 Introduction & Design Specification

In high-speed Polymer Optical Fibre (POF) links, arriving optical power fluctuates heavily due to transmission distance variance (1m to 20m) and connector misalignment. Without automatic regulation, high-power inputs from short patch cords saturate the high-gain transimpedance amplifier (TIA), inducing heavy pulse-width distortion. Conversely, weak signals from lossy 20m reels fail to trip the subsequent digital comparator.

To resolve this, an Automatic Gain Control (AGC) error amplifier module was engineered to stabilize the intermediate receiver voltage based on the criteria in Table 1.

Table 1: AGC Design Performance Metrics

Parameter	Design Target / Boundary
Power Supply Rails	Primary unipolar +5 V DC, dual ± 5 V for op-amp substrate
Input Tracking Range	0 V to 3.3 V DC envelope from peak detector
Target Output Window	Linear, uncompressed sweep from 1.5 V to 3.5 V
Dynamic Stability	Zero closed-loop high-frequency oscillation or ripple
Downstream Target	Direct gate modulation of an enhancement NMOSFET (BS170)

4.2 Circuit Architecture and Operational Stages

The AGC block consists of three functional stages: first-stage isolation/amplification, passive peak detection filtering, and active error integration.

4.2.1 First Amplification Stage

An LT1252 high-speed current-feedback op-amp provides a high-input-impedance active interface to prevent parasitic loading. Configured for non-inverting gain via feedback resistor $R_2 = 2.2 \text{ k}\Omega$ and gain-setting resistor $R_1 = 220 \Omega$ (AC grounded via C_1), the exact theoretical voltage gain (A_{V1}) is:

$$A_{V1} = 1 + \frac{R_2}{R_1} = 1 + \frac{2.2 \text{ k}\Omega}{220 \Omega} = 11$$

Biased via $\pm 5 \text{ V}$ rails to permit unclipped symmetrical swings around a 0 V baseline, the LT1252 amplifies the input AC amplitude parameter ($A_{\text{in}} = 0.01 \text{ V} \rightarrow 0.4 \text{ V}$) up to a robust $\pm 4 \text{ V}$ range. This active scaling lifts low-level signals far above the forward dead-band threshold of the subsequent passive rectification network.

4.2.2 Inter-Stage Peak Detection Filtering

The amplified AC signal is translated into a slowly varying DC envelope via a passive half-wave peak detector comprising switching diode D_1 , bleed resistor $R_3 = 100 \text{ k}\Omega$, and filtering capacitor $C_4 = 1 \text{ nF}$. The parallel time constant ($R_3 \parallel C_4$) is optimized to hold peak potentials across sequential data pulses without premature decay, while stripping the high-frequency carrier component. The resulting ripple-free DC tracking envelope ($0.3 \text{ V} \rightarrow 3.3 \text{ V}$) is routed via series resistor $R_4 = 5 \text{ k}\Omega$ to the downstream error processor.

4.2.3 Active Error Integration and Closed-Loop Regulation

The final stage utilizes an OP37 op-amp configured as an active summing error integrator. Resistors R_5 ($150 \text{ k}\Omega$) and R_7 ($12 \text{ k}\Omega$) form a fixed potential divider that injects a localized baseline bias into the non-inverting node (Pin 3), where it sums with the rectified envelope via $R_4 = 5 \text{ k}\Omega$. Nodal analysis defines the unconstrained static voltage at Pin 3 as:

$$V_{\text{pin3}} = \left(\frac{V_{\text{in}}}{R_4} + \frac{5 \text{ V}}{R_5} \right) \times (R_4 \parallel R_5 \parallel R_7) \approx (0.6897 \times V_{\text{in}}) + 0.1149 \text{ V}$$

With feedback parameters $R_8 = 33 \text{ k}\Omega$ and $R_6 = 10 \text{ k}\Omega$, the stage closed-loop gain is:

$$A_V = 1 + \frac{R_8}{R_6} = 4.3$$

For an independently imposed detector voltage, sweeping V_{in} from 0.3 V to 3.3 V projects an output span from:

$$V_{\text{out_min}} = [(0.6897 \times 0.3 \text{ V}) + 0.1149 \text{ V}] \times 4.3 \approx 1.38 \text{ V}$$

$$V_{\text{out_max}} = [(0.6897 \times 3.3 \text{ V}) + 0.1149 \text{ V}] \times 4.3 \approx 10.28 \text{ V}$$

In active SPICE simulation, however, negative feedback limits the actual steady-state output to a window of **1.4 V to 3.4 V**. As V_{out} climbs, it drives the downstream BS170 gate, initiating channel attenuation that curbs the raw carrier. This loop interaction prevents V_{in} from ever reaching the 3.3 V open-loop maximum, clamping it near ≈ 1.0 V and protecting the stage from the theoretical 10.28 V saturation overshoot.

4.3 Detailed Schematic & System Parameters

The baseline topology of the integrated three-stage AGC error amplifier is illustrated in Figure 2.

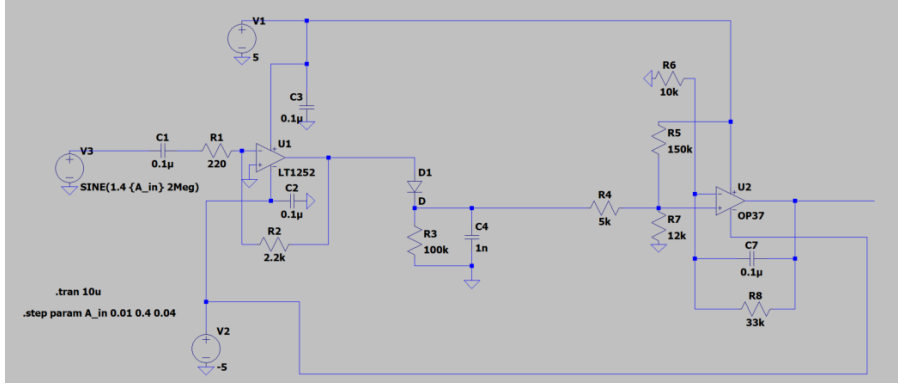


Figure 2: The AGC circuit schematic baseline layout.

The primary network component values for this baseline model are configured as follows: $R_1 = 220\ \Omega$, $R_2 = 2.2\ \text{k}\Omega$, $R_3 = 100\ \text{k}\Omega$, $R_4 = 5\ \text{k}\Omega$, $R_5 = 150\ \text{k}\Omega$, $R_6 = 10\ \text{k}\Omega$, $R_7 = 12\ \text{k}\Omega$, $R_8 = 33\ \text{k}\Omega$, and integration capacitor $C_7 = 0.1\ \mu\text{F}$.

4.4 Simulation Results & Discussion

The closed-loop transient response was verified via LTspice across a stepped input amplitude range. Three operational node waveforms were captured to track signal progression (Figures 3–5).

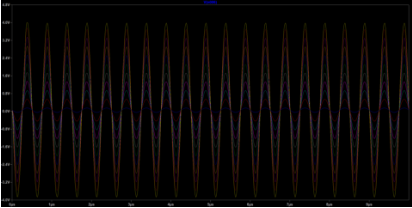


Figure 3: First Stage AC Output

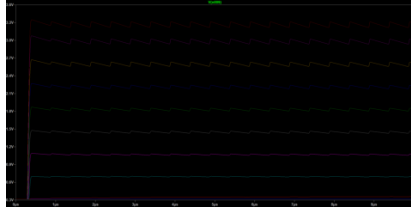


Figure 4: Peak Detector DC Envelope

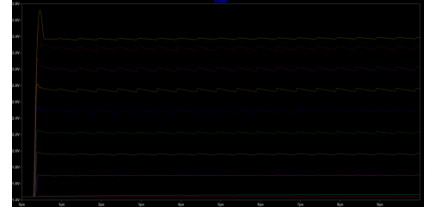


Figure 5: Error Integrator Control Voltage

The first amplification stage tracks the AC input input linearly, generating an unclipped bipolar swing of ± 4 V (Figure 3). The half-wave detector strips high-frequency carrier ripple, outputting a stabilized DC envelope across C_4 (Figure 4).

The final system control response (Figure 5) exhibits a linear staircase profile. Under closed-loop feedback, the minimum floor settles at $V_{\text{out_min}} \approx 1.48$ V, enabling sub-threshold NMOS channel control, while the peak step plateaus smoothly at $V_{\text{out_max}} \approx 3.41$ V. The minor leading-edge startup overshoot (≈ 0.1 V) indicates a responsive loop with rapid settling.

4.5 Final Empirical Saturation Mitigation Tuning

Physical breadboard testing revealed a critical hardware limitation: feeding input envelope voltages above 0.6 V caused the output to flatline at exactly 4.2 V. This observation is consistent with limited

output swing under the tested supply and loading conditions; it does not by itself isolate the internal cause.

To pull the active control envelope away from this 4.2 V lock, the summing bias network was empirically re-scaled and the gain compressed. Pull-up resistor R_5 was increased to 330 k Ω to weaken the +5 V injection, ground-return resistor R_7 was decreased to 10 k Ω to enforce a downward pull on Pin 3, and feedback resistor R_8 was scaled down to 27 k Ω (compressing stage gain to $A_V = 3.7$). This hardware optimization shifted the empirical baseline to 1.5 V and securely capped the peak control ceiling at 3.5 V. High-frequency immunity was reinforced by adding 0.1 μ F decoupling capacitors across the supply pins and a 10 nF RF-damping capacitor from Pin 3 to Ground.

4.6 System Integration & Closed-Loop Interconnection

The optimized control output (1.4 V $\rightarrow$ 3.4 V) of the OP37 block connects directly to the gate of an enhancement-mode BS170 NMOSFET, forming an automated passive shunt-attenuation loop.

The upstream receiver amplifier (AD8055) operates at a substantial fixed voltage gain of $A_{V2} = -\frac{R_{fb}}{R_{in}} \approx -83.3$. This profile ensures sufficient amplification for weak signals from 20m fibre links, but threatens to saturate the receiver when high-power signals from short 1m patch cords are present. Under high-input conditions, the rising peak voltage drives the OP37 output toward its 3.4 V ceiling, forcing the BS170 into its linear region. The channel resistance (R_{DS}) collapses, forming a dynamic potential divider with the series resistor R_{33} (220 Ω):

$$V_{\text{Receiver Out}} = V_{\text{Stage2 Out}} \times \left(\frac{R_{DS}}{R_{33} + R_{DS}} \right)$$

An upward amplitude spike prompts an immediate corrective increase in the AGC output voltage, suppressing R_{DS} and attenuating the AC signal entering the downstream comparator towards the intended operating envelope. This attenuation is intended to reduce excessive signal amplitude at the comparator input. An attenuator downstream of an amplifier cannot recover information already lost through upstream clipping.

5 Comparator and Output Buffer

5.1 Circuit Overview and Adaptive Reference Control

To convert the conditioned analog waveform from the preceding feedforward AGC stage back into a synchronous 2 Mbit/s digital bitstream, a high-speed comparator stage was engineered leveraging the MAX941 architecture. This hardware pipeline addresses the concurrent physical challenges of parasitic threshold chattering, logic-level squaring, and high-current load isolation. The complete structural implementation is presented in Figure 6.

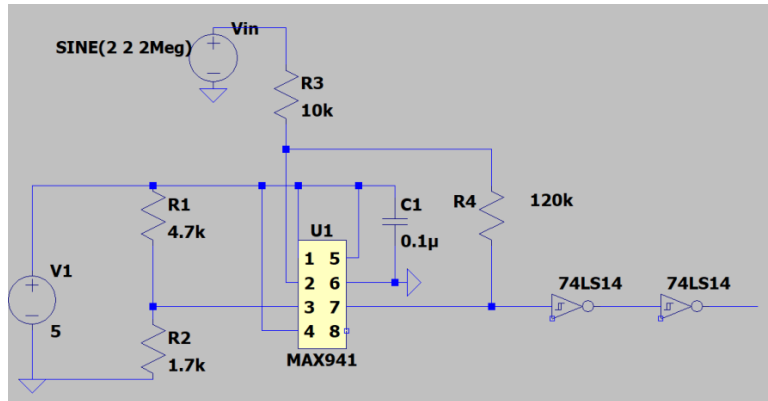


Figure 6: Schematic of the comparator circuit.

The baseline reference voltage V_{ref} is established via a low-impedance resistive network combining a fixed $4.7\text{ k}\Omega$ resistor with a precision $2\text{ k}\Omega$ tuning potentiometer. Rather than deploying high-value resistors that introduce parasitic RC propagation delays and input bias drift, this low-impedance configuration reduces sensitivity to loading. The integrated potentiometer yields manual calibration flexibility to dynamically realign the switching reference against variations in the TIA output baseline. Local power supply isolation is enforced by placing twin $0.1\text{ }\mu\text{F}$ ceramic decoupling capacitors with ultra-short leads directly adjacent to the V_{CC} and GND pins of the MAX941, successfully clamping high-frequency breadboard rail noise.

5.2 Hysteresis Feedback and Empirical Dynamic Verification

To reduce false output bouncing triggered by high-frequency noise transitions at the zero-crossing axes, an asymmetrical positive feedback loop was calculated and introduced to enforce a rigid hysteresis window. A series input resistor $R_3 = 10\text{ k}\Omega$ was mapped to the non-inverting node, coupled with a feedback path of $R_4 = 120\text{ k}\Omega$ bridging the output to the non-inverting input. Applying Kirchhoff’s Current Law (KCL) under worst-case transient logic limits:

$$\frac{V_{\text{in}} - V_{\text{ref}}}{R_3} = \frac{V_{\text{ref}} - V_{\text{out}}}{R_4}$$

Evaluating the system across the binary states ($V_{\text{out, high}} = 5\text{ V}$ and $V_{\text{out, low}} = 0\text{ V}$) establishes the theoretical switching thresholds at $V_{\text{TH}} = 1.613\text{ V}$ and $V_{\text{TL}} = 1.187\text{ V}$, yielding a robust hysteresis noise margin of $\Delta V_{\text{hyst}} \approx 426\text{ mV}$. Empirical characterization executed at a low-frequency sweep baseline proved that the MAX941’s intrinsic 80 ns propagation delay introduces negligible phase-lag, ensuring that the physical switching thresholds are consistent with the derived design equations.

To fulfill the stringent fan-out and edge-rate requirements of high-speed digital receiver logic, the output of the MAX941 (Pin 7) is fed directly into a **74LS14 Schmitt-trigger buffer**. Configured as two consecutive inverters to preserve the native logic polarity, this dedicated digital buffering stage provides additional output drive (I_{out}) and sharpens the transitions into sharper logic transitions, independent of the subsequent loading network.

The dynamic tracking efficiency of this complete digitization framework was verified empirically on the Keysight digital oscilloscope, as illustrated by the synchronized transient traces in Figure 7 and Figure 8.

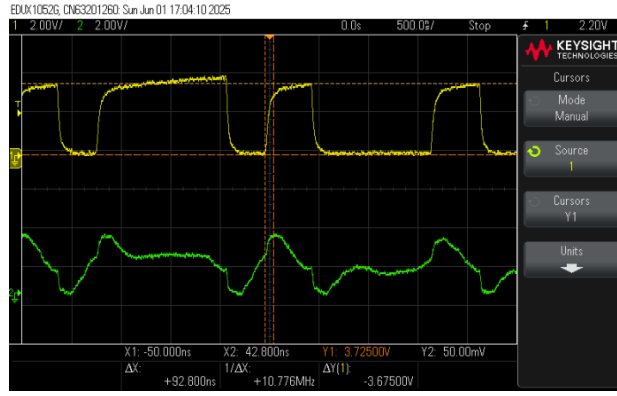


Figure 7: Digital comparator output (yellow) contrasted against the receiver output signal (green).

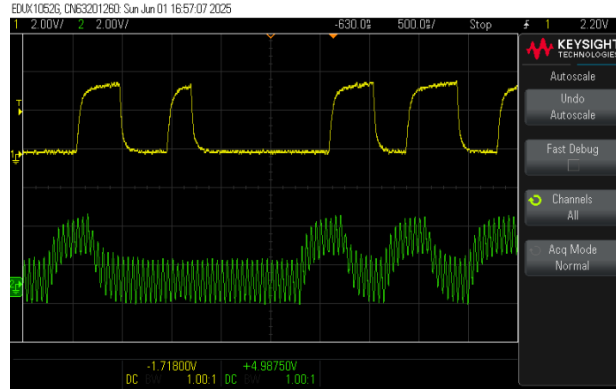


Figure 8: Digital comparator output (yellow) tracked against the transmitter output signal (green).

As captured in Figure 7, the comparator output (yellow trace) tracks the peaks and valleys of the highly non-ideal receiver output (green trace). The calculated ΔV_{hyst} successfully prevents false triggering amidst the significant analog high-frequency ripples, confirming excellent operational noise immunity. Furthermore, Figure 8 demonstrates a near-flawless phase and timing alignment between the final comparator digital output (yellow) and the original transmitter input signal (green). The comparison highlights a pristine temporal correlation with minimal bit-width distortion or jitter, proving that the receiver detects almost the entirety of the high-speed payload successfully.

5.3 Interstage Attenuation and Impedance Isolation Buffer

The squared 0 V to 5 V TTL square wave delivered from the 74LS14 buffer cannot be injected into a physical $8\ \Omega$ speaker load without causing severe loading saturation and excessive thermal dissipation. To safely bridge this digital-to-analog boundary, a passive 10 : 1 potential divider ($R_{\text{div}1} = 900\ \Omega$, $R_{\text{div}2} = 100\ \Omega$) is connected to attenuate the voltage transitions down to a standard 500 mV_{pp} consumer audio line-level signal ($V(n005)$).

Transient simulation in Figure 9 confirms that this attenuated node tracks the 2 Mbit/s switching edges with absolute phase synchronization. Because the $90\ \Omega$ Thévenin output impedance of the resistive divider is far too weak to drive the heavy speaker, an LT1252 current-feedback operational amplifier—configured as a unity-gain voltage follower—was deployed by the author to provide load impedance isolation. Boasting a robust 150 mA driving capacity, the LT1252 isolates the sensitive logic stages from the load, while a downstream high-pass AC coupling network ($R_4 = 50\ \Omega$, $C_1 = 100\ \text{nF}$) filters out remaining DC bias, delivering uncompromised audio power to the destination.

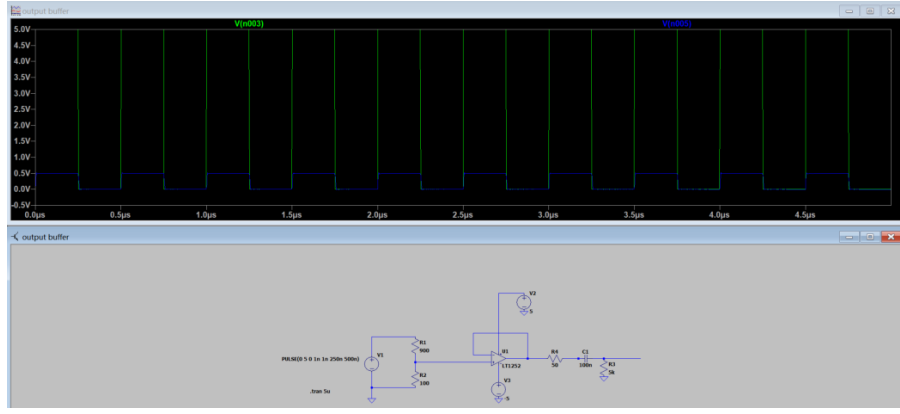


Figure 9: LTspice simulation of the output stage, contrasting the 5 V pre-attenuated comparator output $V(n003)$ with the 500 mV buffered audio drive $V(n005)$.

6 Experimental Characterisation and Signal Integrity

To evaluate the signal integrity of the electro-optical link operating at the target data rate of 2 Mbit/s, eye-diagram measurements were captured on the receiver signal path using a Keysight digital oscilloscope. The resulting eye diagram provides a visual indication of the system noise margin, timing stability and susceptibility to inter-symbol interference (ISI).

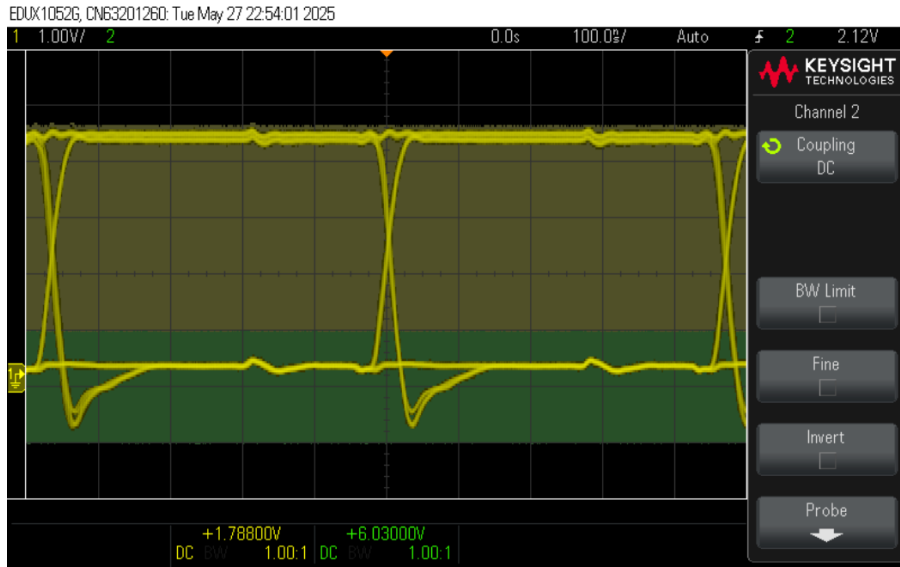


Figure 10: Measured receiver eye diagram at 2 Mbit/s, displaying a large eye opening and transient undershoot following signal transitions.

Quantitative analysis of the measured eye diagram yields two key performance metrics governing the reliability of the communication link:

- Voltage Margin (Vertical Eye Opening):** The eye diagram exhibits a large vertical opening, indicating a substantial voltage margin between the logical high and low states. The logical high level remains well defined throughout the sampling interval, while the overall eye opening spans several volts. A transient undershoot and subsequent ringing are observed immediately following the falling edge, reaching approximately 1 V below the steady-state low level before settling within approximately 100 ns. Despite this behaviour, the central sampling region remains largely unaffected. The large eye opening suggests good noise immunity and provides significant margin against threshold detection errors, reducing the likelihood of false switching caused by transient noise.

- **Timing Jitter (Horizontal Eye Closure):** The crossing region of the eye remains relatively narrow, indicating low timing jitter. The crossing thickness was estimated to be approximately $\Delta t_{\text{jitter}} \approx 30$ ns. For a data rate of 2 Mbit/s, the corresponding bit period is $T_{\text{bit}} = \frac{1}{2 \times 10^6} = 500$ ns. The measured jitter therefore represents approximately 6% of the bit interval. This relatively small horizontal closure indicates good timing stability throughout the link. No significant dispersion-related broadening was observed over the 20 m polymer optical fibre channel.

Overall, the measured eye diagram remained clearly open at the target data rate, demonstrating sufficient voltage and timing margin for reliable digital data recovery across the optical communication link.

7 Conclusion

The end-to-end electro-optical communication link engineered by the **Bandwidth Bandits** successfully met the primary performance objectives outlined in the project brief. Reliable synchronous digital data transmission at the target rate of 2 Mbit/s was achieved and characterised over a 20 m polymer optical fibre (POF) channel using a red LED source ($\lambda = 650$ nm).

Through a structured division of labour and progressive system integration, the team successfully transformed the initial system concept into a fully functional hardware prototype. Group-level characterisation established an overall channel attenuation of approximately 13 dB, while the individual subsystem developments provided the transmitter, receiver, comparator, AGC and testing infrastructure required for reliable link operation.

The receiver chain successfully recovered transmitted data under varying optical power conditions. The AGC architecture was developed to improve tolerance to signal amplitude variations and reduce the likelihood of receiver under-drive and amplifier saturation. Eye-diagram measurements demonstrated a large vertical eye opening and low timing jitter ($\Delta t_{\text{jitter}} \approx 30$ ns), indicating good noise margin and stable operation at the target data rate. The demonstration and eye diagram support operation under the tested conditions. A quantified bit-error-rate measurement and a calibrated attenuation sweep would be needed to establish the error rate and sensitivity limit.

The project highlighted the importance of iterative debugging and system integration. Although the initial design phase suggested a relatively straightforward implementation, considerable effort was required to resolve parasitic oscillation, grounding issues, power-supply coupling effects and interface mismatches between independently developed subsystems. In practice, system integration proved significantly more challenging than the initial circuit design process.

Finally, the successful transmission of a continuous CD audio data stream to a loudspeaker demonstrated the practical functionality of the completed optical communication system and provided additional validation of its signal integrity and overall system performance. The project provided valuable experience in high-speed analogue circuit design, optical communication systems, signal integrity analysis and collaborative engineering development.

A Sub-system Circuit Diagrams

To provide granular, component-level documentation for the end-to-end communication link, this section compiles the complete set of localized circuit diagrams engineered by the **Bandwidth Bandits** design group. These schematics formally substantiate the hardware implementation stages outlined in the main text.

A.1 Transceiver Core System Infrastructure

Figure 11 and Figure 12 jointly represent the complete hardware circuit topology of the functional communication link. Figure 11 details the electrical configuration of the optical transmitter stage, highlighting the high-speed LED modulation driver. Figure 12 displays the high-gain optical receiver front-end layout designed to capture the attenuated photonic payload from the physical fibre channel.

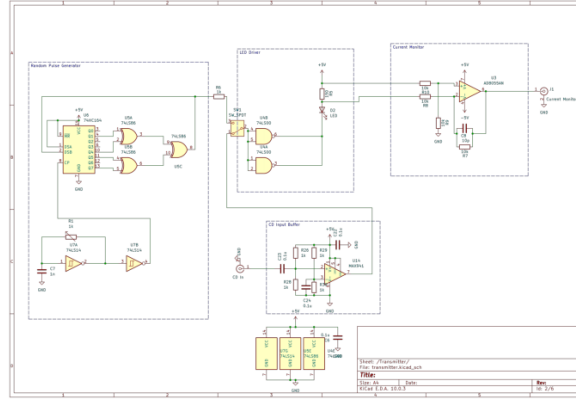


Figure 11: Transmitter circuit diagram detailing the high-speed LED driver stage.

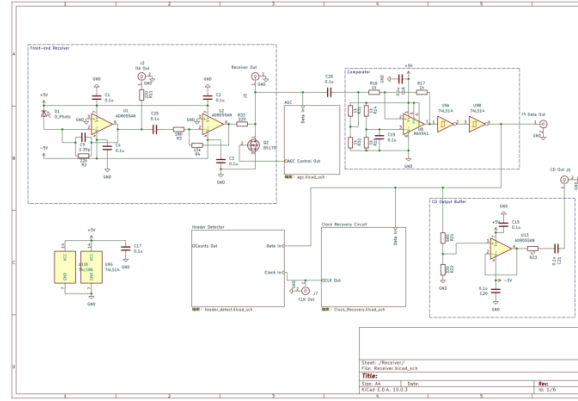


Figure 12: Receiver front-end circuit schematic highlighting the photodiode interface.

A.2 Localized Receiver Internal Sub-circuits

To provide absolute clarity on the digital recovery pipeline, the discrete sub-circuits that are nested within the receiver system but omitted from the main receiver layout due to formatting limits are documented below. Figure 13 details the resistive feedforward Automatic Gain Control (AGC) block. Figure 14 captures the logic gate infrastructure of the hardware-level header detection block, and Figure 15 records the complete discrete loop implementation of the synchronous clock recovery stage.

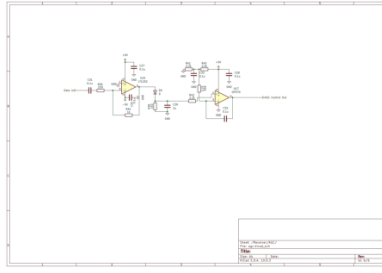


Figure 13: Detailed schematic of the feedforward AGC signal conditioning block.

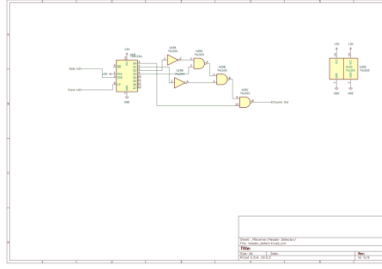


Figure 14: Circuit schematic of the hardware-level header detection logic.

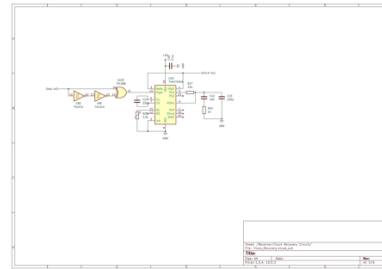


Figure 15: Circuit schematic of the synchronous clock recovery loop block.